

Master's Thesis

Evaluating foundation model embeddings in RNA-Protein Interaction classification with RPIembeddor

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Declaration

I hereby declare, that I am the sole author and composer of my thesis and that no other sources or learning aids, other than those listed, have been used. Furthermore, I declare that I have acknowledged the work of others by providing detailed references of said work.

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Abstract

RNA–protein interactions (RPIs) play a fundamental role in numerous cellular processes, including gene regulation and RNA metabolism. Experimental identification of RPIs is often expensive and time-consuming, motivating the development of computational methods for interaction prediction. Recent advances in biological foundation models have enabled the generation of rich sequence embeddings that can be transferred to downstream bioinformatics tasks.

In this work, we investigate the integration of embeddings from different biological foundation models into RPIembeddor, a Transformer-based model for RNA–protein interaction prediction operating on frozen embeddings. We first reproduced the original RPIembeddor framework and its RNAInterAct dataset, achieving comparable and, in some settings, improved performance. Hyperparameter optimization was subsequently performed using NePS with the Hyperband search strategy.

To study the influence of different embedding models, we evaluated three additional protein foundation models and two additional RNA foundation models on both the original RNAInterAct dataset and a newly constructed eCLIP-based dataset with alternative negative generation strategies. We evaluated embedding quality using unsupervised representation metrics.

Our results show that the choice of foundation model significantly affects downstream performance, although no single model consistently outperformed all others across datasets and evaluation settings. ESM-2 showed robust performance on the protein side, while RNA model performance was dataset-dependent, with LAMAR and RNAElectra excelling under different conditions. Furthermore, unsupervised embedding metrics were only partially predictive of downstream performance, highlighting the complex relationship between intrinsic representation quality and RPI prediction.

Overall, our findings suggest that improvements in RNA–protein interaction prediction depend not only on the choice of foundation models, but also on dataset construction, negative sampling strategies, and evaluation protocols. These results contribute to a better understanding of the role of biological foundation models in RPI

prediction and provide insights for the development of more robust and generalizable computational approaches.

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Views and opinions expressed are those of the author only.

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1 Introduction

Several biological processes are driven by molecules that work together in a coordinated manner. In particular, RNA and proteins are deeply intertwined and participate in many cellular mechanisms that allow life to function.

One way to shed light on these processes is to determine whether a given RNA and protein pair interact. Such interactions are rarely arbitrary; when they occur, they usually serve a biological purpose [3].

In most cases, the most reliable way to identify these interactions is through wet-lab experiments. However, performing such experiments is often time-consuming, expensive, and limited in throughput.

Computational methods have steadily gained reliability over recent decades and are therefore attractive due to their ability to generate predictions more quickly and at a lower cost.

Computational approaches for RNA–protein interaction (RPI) prediction include statistical models, structure-based methods, biophysical modeling, supervised learning, and more recently, self-supervised learning approaches [4, 5, 6, 7].

Self-supervised methods have become particularly valuable due to the availability of large models trained on massive biological sequence datasets. These models are capable of encoding meaningful information about the molecules they are trained on and are commonly referred to as Foundation Models (FMs). Several RNA and protein foundation models have been proposed and represent promising candidates for RPI prediction, including RNA-FM [8], LAMAR [9], RNA-ELECTRA [10], ESM-2 [11], ESM-C [12], VESM [13], and LC-PLM [14].

Previous work has shown that the representations generated by these foundation models can be successfully transferred to a variety of downstream biological tasks, including RNA–protein interaction prediction.

RPIembeddor [15] uses only sequence information from RNAs and proteins. The model employs self-attention and cross-attention mechanisms to model interactions between residues from both molecules. By operating solely on sequence information, the method abstracts away many structural details and focuses on the residue-wise

relationships encoded in the sequences.

The main motivation of this thesis is therefore to investigate whether alternative foundation models can improve the generalization capabilities of the RPIembeddor framework.

The main objective of this thesis is to reproduce the original RPIembeddor framework and evaluate the effect of different RNA and protein foundation models on RNA–protein interaction prediction. Additional experiments investigate alternative datasets, and adaptations required to support different embedding models within the RPIembeddor pipeline.

To clarify this improvement in generalization, we aim to answer the following research questions:

RQ1: Is there a difference in performance between different protein FMs embeddings and RNA FMs embeddings combinations?

RQ2: Can training on larger and more biologically meaningful datasets improve the generalization of the model?

RQ3: Are the pretrained representations of these models truly meaningful when compared against simpler approaches such as feed-forward neural networks?

The contributions of this work are summarized as follows:

- Generation of frozen embeddings from a selection of relevant RNA and protein foundation models, followed by their evaluation using unsupervised embedding quality metrics and downstream RPIembeddor performance. This addresses RQ1.
- Investigation of training on larger datasets obtained from protocols containing richer RPI signal in order to evaluate the effect of dataset scale and quality on model generalization. This addresses RQ2.
- Comparison of pretrained foundation models against simpler baselines, including a CNN, in order to assess the importance of pretraining. This addresses RQ3.

The implementation developed for this thesis, including the training pipeline, dataset generation scripts, and evaluation code, is publicly available at:

<https://github.com/josgiort/RPIEmbeddor2.0>

For the purposes of this work, RNAs and proteins are represented solely as sequences. Although RNA–protein interactions occur between complex three-dimensional molecular structures, the methods investigated here operate only on sequence information. An example of an RPI complex is shown in figure below (PDB entry 1G2E), where the protein chain is shown as a uniform tube colored by amino acid type (blue-green palette), and the RNA chain is shown as a flat ribbon colored by nucleotide identity (A, U, G, C; red-orange palette). The structures are simplified to highlight the residue-level abstraction used throughout this thesis.

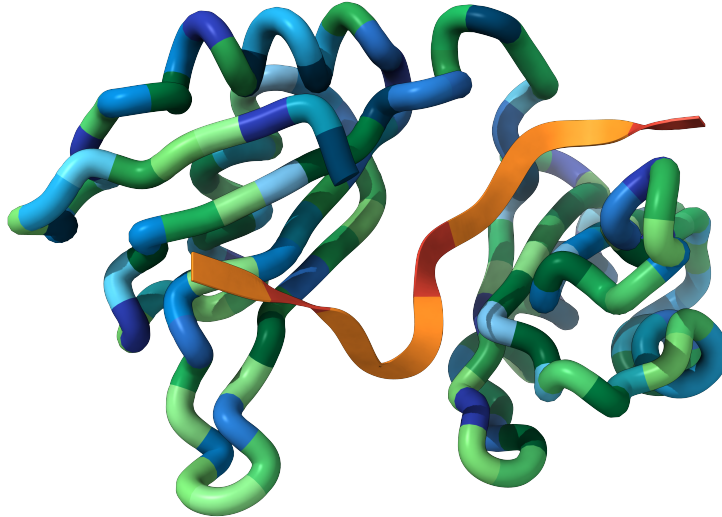


Figure 1: Abstraction of sequences highlighting only residues. Modeled using AlphaFold 3 [2] and visualized with UCSF ChimeraX [1].

2 Background

2.1 Biological Background

2.1.1 RNA

The Ribonucleic Acid is a single-stranded nucleic acid composed of ribonucleotides containing the bases adenine (A), uracil (U), cytosine (C), and guanine (G). RNA is involved in the storage, transfer, and regulation of genetic information, particularly in protein synthesis and gene regulation.[16]

2.1.2 Proteins

A Protein is a biological macromolecule made of amino acid chains linked by peptide bonds. Proteins perform essential cellular functions, including catalysis, transport, structural support, signaling, and molecular binding.[16]

2.1.3 RNA - Protein Interactions

RNA-protein interactions are fundamental to a wide range of biological processes, including gene expression, post-transcriptional regulation, and RNA processing [3]. Proteins interact with RNA molecules through specific recognition of nucleotide sequences and structural motifs, forming ribonucleoprotein (RNP) complexes that are essential for cellular function [3]. The structural characterization of these complexes is critical for understanding the molecular basis of such interactions. Recent advances in computational structural prediction, particularly with AlphaFold 3, have enabled the accurate modeling of protein-nucleic acid complexes [2], opening new possibilities for studying RNA-protein recognition at atomic resolution.

Accurately identifying whether a given RNA and protein interact is therefore of considerable biological interest. Computational prediction of RNA-protein interactions can help prioritize experimental validation efforts, improve our understanding of cellular regulatory mechanisms, and facilitate the discovery of novel functional interactions [17].

2.1.4 eCLIP

Enhanced Crosslinking and Immunoprecipitation (eCLIP) is a high-throughput experimental technique used to identify RNA–protein interactions in vivo. It combines ultraviolet crosslinking, immunoprecipitation of RNA-binding proteins, and high-throughput sequencing to determine the RNA binding sites of proteins across the transcriptome with improved efficiency and reduced background noise compared to earlier CLIP methods. [18]

2.2 Transformers and Foundation Models

2.2.1 Transformer Architecture

The Transformer is a neural network architecture based entirely on attention mechanisms, enabling parallel processing of sequential data without recurrent structures. Introduced by Vaswani et al. [19], the Transformer models dependencies between sequence elements using self-attention.

Given an input sequence $X = [x_1, x_2, \dots, x_n]$, the Transformer computes three matrices:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

where Q , K , and V denote the query, key, and value matrices, and W_Q , W_K , and W_V are learnable parameter matrices.

The scaled dot-product attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

where d_k is the dimensionality of the key vectors.

To capture information from multiple representation subspaces, the Transformer employs multi-head attention:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W_O$$

with

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

where h is the number of attention heads and W_O is a learnable projection matrix.

Since the Transformer contains no recurrent structure, positional encodings are added to input embeddings to preserve sequence order information:

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d}}\right)$$

$$PE(pos, 2i + 1) = \cos\left(\frac{pos}{10000^{2i/d}}\right)$$

where pos represents the token position and d is the embedding dimension.

2.2.2 ESM Transformer Architecture

The **Evolutionary Scale Modeling (ESM)** Transformer is a protein language model based on the Transformer architecture, designed to learn representations of protein sequences using self-supervised learning on large-scale biological datasets. ESM models are trained on millions of protein sequences using masked language modeling, enabling the model to capture structural, functional, and evolutionary information from amino acid sequences.

Given a protein sequence

$$X = [x_1, x_2, \dots, x_n]$$

where each x_i represents an amino acid token, the sequence is first embedded into a continuous vector space and combined with positional embeddings:

$$H^{(0)} = E(X) + P$$

where $E(X)$ denotes the amino acid embedding matrix and P represents positional encodings.

The embedded sequence is processed through multiple Transformer encoder layers composed of multi-head self-attention and feed-forward networks. The self-attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

where

$$Q = HW_Q, \quad K = HW_K, \quad V = HW_V$$

and W_Q , W_K , and W_V are learnable projection matrices.

The hidden representation at layer l is computed as:

$$H^{(l+1)} = \text{TransformerLayer}(H^{(l)})$$

After passing through all encoder layers, the final hidden representations encode contextual biochemical and evolutionary information for each amino acid residue. These representations can be used for downstream tasks such as protein function prediction, structure prediction, and RNA–protein interaction classification.

2.3 Biological Language Models

2.3.1 Foundation Model

Foundation model is a large-scale machine learning model typically trained on vast amounts of unlabeled data using self-supervised learning. Foundation models learn general-purpose representations that can be adapted to multiple downstream tasks, such as classification, prediction, generation, or sequence analysis, through fine-tuning or transfer learning. [20] In bioinformatics, models such as protein language models use this approach to capture structural and functional information from biological sequences. [11, 8].

2.3.2 RNA Models

Recent advances in biological foundation models have led to the development of large language models specifically designed for RNA sequences. These models are typically trained using self-supervised objectives on large collections of RNA sequences and aim to learn representations that capture structural and functional properties of RNA molecules.

RNA-FM RNA-FM is one of the first large-scale RNA foundation models trained on more than 23 million non-coding RNA sequences using a masked language modeling objective. The model follows the Transformer architecture and has demonstrated strong performance on RNA structure prediction and several downstream functional tasks. Due to its ability to generate contextual residue-level embeddings, RNA-FM has become a commonly used backbone for RNA representation learning [8]. This was the RNA Foundation Model used by RPIembeddor.

LAMAR LAMAR is a large-scale RNA foundation model designed to capture diverse regulatory mechanisms encoded within RNA sequences. The model was trained on

large transcriptomic datasets and aims to provide transferable representations for a broad range of downstream RNA prediction tasks, including regulatory inference and transcript-level analyses [9].

RNA-ELECTRA RNA-ELECTRA adapts the ELECTRA pretraining framework to RNA sequences. Instead of reconstructing masked tokens, the model is trained using replaced-token detection, allowing it to learn informative sequence representations in a computationally efficient manner. The resulting embeddings can be transferred to various downstream RNA analysis tasks [10].

2.3.3 Protein Models

Protein language models apply self-supervised learning techniques to large protein sequence databases in order to learn representations that capture evolutionary, structural, and functional information. Similar to language models in natural language processing, these models learn contextual representations directly from amino acid sequences.

ESM-2 ESM-2 is a family of Transformer-based protein language models trained using masked language modeling on millions of protein sequences. Scaling from millions to billions of parameters, ESM-2 has demonstrated strong performance across numerous protein prediction tasks and serves as the backbone of several state-of-the-art protein structure and function prediction systems [11]. This was the protein Foundation Model used by RPIembeddor.

ESM-C ESM-C is a more recent member of the ESM family designed to provide competitive protein representations at reduced computational cost. The model retains the Transformer-based architecture while improving inference efficiency, making it attractive for large-scale embedding generation and downstream biological prediction tasks [12].

VESM VESM is a compressed protein language model that distills the collective knowledge of multiple ESM models into a single architecture. The objective is to retain the representational quality of larger protein language models while reducing computational requirements. VESM has shown strong performance on variant effect prediction and related downstream protein tasks [13].

LC-PLM LC-PLM replaces the Transformer architecture with a bidirectional Mamba-based architecture built upon selective state-space models. Unlike standard Transformers, whose memory requirements scale quadratically with sequence length, LC-PLM scales approximately linearly, enabling the processing of substantially longer protein sequences. This property makes it particularly attractive for tasks involving large proteins and long-range sequence dependencies [14].

On the following table, relevant information regarding each foundation model can be seen.

Table 1: Foundation models used for embedding generation. The table reports the modality, model version, architecture, model size, hidden representation dimensionality, and maximum context length used during embedding extraction.

Model	Modality	Version	Architecture	Model Size	Hidden Dim.	Context Length
RNA-FM	RNA	t12	Transformer	150M	640	1022
LAMAR	RNA	150M	Transformer	150M	768	1024
RNAElectra	RNA	Base	Transformer	93M	512	1022
ESM-2	Protein	t30	Transformer	150M	640	1022
ESMC	Protein	600M	Transformer	600M	1152	2046
VESM	Protein	150M	Transformer	150M	640	1022
LC-PLM	Protein	Base	BiMamba	–	1536	>16k

2.4 RPIembeddor

2.4.1 Model

RPIembeddor [15] is a Transformer-based deep learning model developed for **RNA–protein interaction (RPI) binary classification**. The model predicts whether an RNA and a protein interact by leveraging pretrained sequence embeddings generated from biological foundation models.

RPIembeddor uses separate pretrained embedding models for RNA and protein sequences to capture structural, functional, and evolutionary information. These embeddings are combined and processed through Transformer encoder layers and classification modules to learn interaction patterns between RNA and protein representations.

The model was introduced together with the *RNAInterAct* dataset and demonstrated strong generalization performance across multiple benchmark datasets for ncRNA–protein interaction prediction.

2.4.2 Architecture

The architecture consists of three main stages:

Embedding Generation RNA and protein sequences are first encoded using pretrained foundation models. RNA representations are generated using RNA Foundation Model (RNA-FM), while protein representations are extracted using Evolutionary Scale Modeling (ESM-2). Both models produce contextual embeddings of dimension $N \times 640$, where N represents the sequence length. These embeddings remain frozen during downstream training.

The embedding generation needs to be carried out before use of RPIembeddor, as the input for the model are actually the embeddings of unique sequences present in the train, validation and test datasets.

The inputs are a parquet file containing all unique RNA sequences and a parquet file containing all unique protein sequences. Then there is a call to the respective embedding model to generate the embeddings sequence by sequence (embedding generation is done separately for RNA and protein sequences). More specifically, to generate the embeddings, first a tokenizer receives the raw sequences and converts them into the required input format for the respective foundation model. The embeddings are created by doing inference on the encoder of the given foundation model (last hidden state) as the representations of interest. Every sequence embedding is saved as a numpy array file with BOS (Beginning of Sequence) and EOS (End of Sequence) tokens remove. Therefore only residue token representations. Final embeddings then have the shape (Number of tokens, Hidden dimension). In the following figure, an overview of the embedding generation routine is presented.

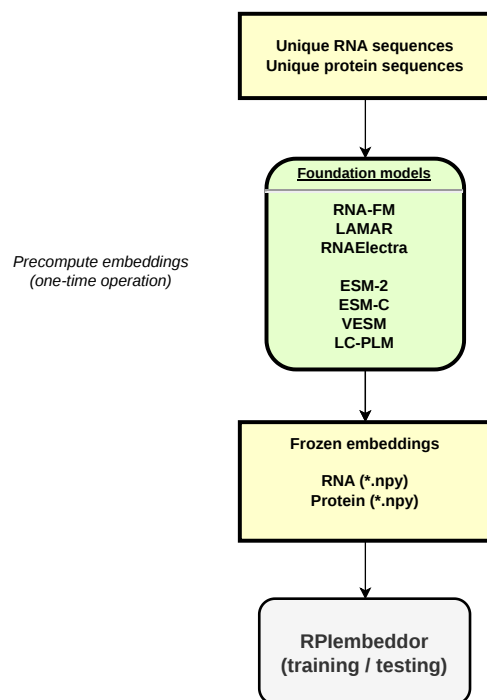


Figure 2: Embedding generation overview.

Transformer Interaction Module The RNA and protein embeddings are passed through parallel feed-forward layers to normalize their dimensions and ensure compatibility. The normalized representations are then processed symmetrically through a Transformer encoder layer with:

- embedding dimension $d_{\text{model}}=256$
- multi-head attention with $n_{\text{head}}=2$,
- a self-attention mechanism for capturing long-range dependencies and interaction patterns between sequences.

This symmetrical attention-based processing allows the model to dynamically focus on the most relevant regions of both RNA and protein sequences for interaction prediction.

Classification Head The latent RNA and protein representations are concatenated and passed through a deep feed-forward network composed of 20 layers. The final prediction is produced by a linear layer followed by a sigmoid activation function, outputting the probability of interaction between the RNA–protein pair.

The complete architecture contains approximately 1.4 million trainable parameters. Training is performed using binary classification loss optimized with the AdamW optimizer, together with linear warm-up and cosine annealing learning-rate scheduling. Regularization techniques include dropout and weight decay to improve generalization performance. The following figure shows the RPIembeddor architecture.

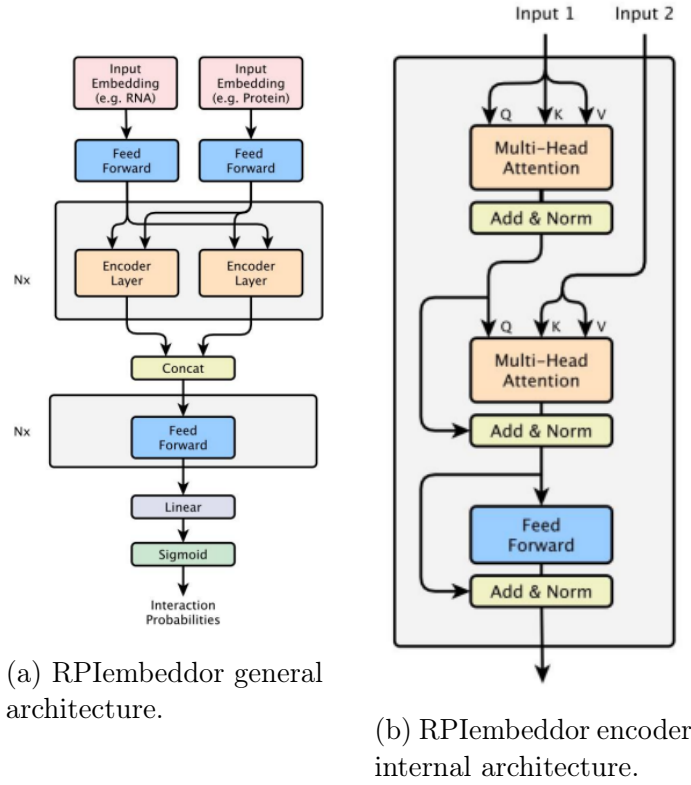


Figure 3: Overview of the RPIembeddor architecture for RNA-protein interaction classification.

2.4.3 Frozen embeddings

Frozen Embeddings are pretrained vector representations whose parameters remain fixed during downstream model training. Instead of updating the embedding model weights, the generated embeddings are used as static input features for another model, reducing computational cost and preserving the knowledge learned during pretraining. In biological sequence modeling, frozen embeddings from pretrained Transformer models are commonly used to represent RNA or protein sequences for classification tasks.

2.4.4 Downstream task

In the context of the Transformer-based model RPIembeddor, the downstream task refers to the specific application for which pretrained RNA and protein representations are adapted and optimized. Here, the downstream task is **RNA-protein interaction (RPI) binary classification**, where the model predicts whether a given RNA-protein

pair interacts (1) or does not interact (0).

The pretrained embeddings extracted from foundation models are used as input features, and the model is trained to learn interaction patterns between RNA and protein sequences for accurate classification.

2.5 Unsupervised Evaluation Metrics for Embeddings

Unsupervised evaluation metrics assess the representational quality of embeddings without relying on labeled downstream tasks. Instead, they analyze properties of the embedding space itself, such as effective dimensionality, information distribution, and embedding isotropy [21].

2.5.1 RankMe

RankMe estimates the effective rank of an embedding matrix by measuring the entropy of its singular value distribution. Given an embedding matrix X with singular values σ_i , the normalized singular values are

$$p_i = \frac{\sigma_i}{\sum_{j=1}^r \sigma_j},$$

where r denotes the rank of X . The RankMe score is then defined as

$$\text{RankMe}(X) = \exp \left(- \sum_{i=1}^r p_i \log p_i \right).$$

Higher RankMe values indicate richer and less redundant representations.

2.5.2 NESum

NESum evaluates how evenly information is distributed across the eigenspectrum of the embedding covariance matrix. Let λ_i denote the covariance eigenvalues. NESum is defined as

$$\text{NESum}(X) = \frac{\left(\sum_{i=1}^d \sqrt{\lambda_i} \right)^2}{\sum_{i=1}^d \lambda_i},$$

where d is the embedding dimension. Larger values indicate a more balanced utilization of the embedding space and reduced dimensional collapse.

2.5.3 Stable Rank

Stable Rank provides a robust estimate of the effective dimensionality of an embedding matrix. It is computed as

$$\text{StableRank}(X) = \frac{\|X\|_F^2}{\|X\|_2^2},$$

where $\|X\|_F$ and $\|X\|_2$ denote the Frobenius and spectral norms, respectively. Higher Stable Rank values suggest that information is distributed more uniformly across embedding dimensions.

2.5.4 IsoScore

IsoScore measures how uniformly an embedding space utilizes its available dimensions [22]. Unlike rank-based metrics, which focus on the effective dimensionality of the embeddings, IsoScore quantifies the degree of isotropy of the embedding distribution.

The metric evaluates whether the variance of the embeddings is distributed evenly across dimensions or concentrated in a small subset of directions. Embeddings with highly concentrated variance are considered anisotropic and may indicate dimensional collapse, whereas isotropic embeddings make more effective use of the available representational space.

IsoScore is normalized to the interval $[0, 1]$, where values close to 1 indicate highly isotropic representations and values close to 0 indicate strong anisotropy.

Formally, IsoScore is computed by comparing the normalized covariance eigenspectrum of the embeddings against an ideal isotropic distribution. Higher IsoScore values therefore indicate a more uniform utilization of the embedding space and a lower degree of representational collapse.

The steps to compute IsoScore are presented in the following algorithm.

Algorithm 1 Computation of IsoScore.

Require: Embedding matrix $X \in \mathbb{R}^{n \times d}$

Ensure: IsoScore value in $[0, 1]$

- 1: Center the embeddings by subtracting the mean of each feature.
 - 2: Apply PCA to rotate the embeddings into principal component space.
 - 3: Compute the covariance matrix of the transformed embeddings.
 - 4: Extract the variance along each principal component.
 - 5: Normalize the variance vector to remove scale effects.
 - 6: Measure the deviation of the normalized variance distribution from an ideal isotropic distribution.
 - 7: Convert the deviation score into a normalized isotropy score.
 - 8: **return** IsoScore
-

3 Related Work

RNA–protein interactions (RPIs) play a crucial role in gene expression regulation and many other cellular processes. As a result, a large number of computational approaches have been proposed for RPI prediction.

3.1 Classical Computational Approaches

Early approaches relied heavily on biological prior knowledge and handcrafted representations. Template-based methods used homology information to infer interactions and were particularly effective when suitable homologous sequences were available. Representative examples include DBD-Threader [23] and SPOT-Seq [24]. However, such approaches are limited by the availability of homologous sequences.

Graph-based methods have also been proposed to model interaction networks and exploit topological information [25, 26, 27, 28]. These methods typically leverage interaction graphs and similarity networks to infer novel interactions.

Other approaches employed traditional machine learning models such as Support Vector Machines and Random Forests. Examples include RPI-Seq [4] and IncPro [5].

Most of these methods relied on handcrafted feature representations derived from RNA and protein sequences. Common representations included one-hot encodings, k-mer frequencies, structural descriptors, and amino acid composition features [29]. While effective, the construction of such features is often labor-intensive and requires substantial domain expertise.

3.2 Deep Learning Methods for RPI Prediction

Deep learning approaches introduced representation learning directly into the prediction pipeline. Methods such as IPMiner [6] and RPI-SAN [7] employed neural architectures to learn task-specific representations from sequence-derived features.

More recent approaches have explored increasingly sophisticated architectures. For example, BioLLMNet [30] performs complex transformations of pretrained embeddings before downstream prediction.

Another notable example is BERT-RBP [31], which fine-tunes a transformer-based model on eCLIP data. Unlike joint RNA–protein interaction models, BERT-RBP focuses on protein-specific prediction tasks and trains a separate model for each protein.

3.3 Foundation Models for Genomics and Proteomics

The success of large language models in natural language processing has motivated the development of biological foundation models trained on large-scale sequence datasets. These models aim to learn general-purpose representations that can be transferred to a wide variety of downstream tasks.

Examples include protein language models such as ProLLaMA [32], ESM-2 [11], ESM-C [12], and LC-PLM [14], as well as RNA foundation models such as RNA-FM [8], LAMAR [9], and RNA-ELECTRA [10].

However, the benefits of large-scale pretraining are not always unequivocal. Several studies have investigated the representational quality of biological foundation models and observed that improvements can be highly task-dependent [33]. Similarly, studies suggest that, for some downstream tasks, pretrained models may not consistently outperform randomly initialized alternatives when used as frozen feature extractors [34].

3.4 Datasets and Benchmarking

Several databases containing experimentally validated RNA–protein interactions have been developed, including eCLIP [18], CLIPdb [35], DoRiNA 2.0 [36], and POSTAR [37].

The construction of high-quality datasets remains a challenging aspect of RPI prediction. While positive interaction examples are generally available from experimental studies, negative examples are rarely observed directly and must often be generated through biologically motivated heuristics [17, 38].

3.5 Foundation Model Evaluation and Embedding Quality

Large-scale benchmark studies have recently emerged to evaluate foundation models across diverse biological tasks. For RNA-related applications, MRNABench provides a curated benchmark covering multiple prediction tasks and evaluation protocols

[39]. The benchmark primarily relies on linear probing of frozen embeddings to assess representation quality.

Similarly, PFMBench [40] evaluates a large collection of protein foundation models across multiple downstream tasks and fine-tuning settings.

Besides downstream task performance, several methodologies have been proposed to directly assess embedding quality. In particular, unsupervised evaluation metrics are attractive because they are computationally inexpensive and can be computed without task-specific labels [22, 21].

4 Methods

4.1 Problem Definition

In this work, the RPI prediction task is formulated as a binary classification problem. Given an RNA sequence r and a protein sequence p , the objective is to predict whether the two molecules interact. The prediction target can therefore be represented as:

$$f : (r, p) \mapsto y$$

where $y \in 0, 1$ denotes the interaction label, with 1 indicating an interaction and 0 indicating no interaction.

4.2 Overview of the Approach

The work started with the reproduction of the original RPIembeddor reported performance with its original data. Then a new dataset was introduced to see how RPIembeddor handled a different and more diverse kind of data. Later, we introduced new foundation models. We benchmarked combinations of foundation models with respect to RPIembeddor test performance metrics. During this part, we also evaluated embeddings’ representation quality with unsupervised metrics. Finally we did a small ablation experiment to observe if embeddings are or not better inputs than one hot representations for our downstream task. The performance metrics used are the following.

- Binary AUPR
- Binary AUROC
- Binary Accuracy
- Binary F1 Score
- Binary Precision

- Binary Recall

The following sections describe the datasets, model architectures and relevant methodologies used throughout this work, but results are not presented until the next chapter.

4.3 Datasets

4.3.1 RNAInterAct

Description of RNAInterAct data is presented in the respective tables below. Pairs refer to RPI examples (RNA - protein pairs).

Table 2: Summary statistics of the RNAInterAct dataset.

Property	Value
<i>General</i>	
Total RNA–protein pairs	122,199
Positive interactions	40,733 (33.3%)
Negative interactions	81,466 (66.7%)
<i>RNA sequences</i>	
Unique RNA sequences	4,166
Unique RNA families	1,148
Min length	46 nt
Max length	1,004 nt
Mean length	134.2 nt
Median length	106.0 nt
Std length	112.8 nt
<i>Protein sequences</i>	
Unique protein sequences	1,306
Unique protein clans	152
Min length	51 aa
Max length	1,020 aa
Mean length	490.9 aa
Median length	449.5 aa
Std length	225.2 aa
<i>Pairs per protein</i>	
Min	2
Max	591
Mean	93.6
Median	47.5
Std	108.1

Table 3: Dataset split statistics for the RNAInterAct dataset.

Property	Train	Validation	Test
Total pairs	83,160	22,640	16,399
Positive pairs	28,823	5,597	6,313
Negative pairs	54,337	17,043	10,086
% of dataset	68.05%	18.53%	13.42%
Unique RNA families	804	172	172

Validation and test sets contain RNA families not present in the training set,

following the family-disjoint evaluation protocol introduced with RNAInterAct.

4.3.2 eCLIP dataset

RPI negatives were artificially generated with the heuristic present in Results and Discussion chapter. With this data, two splits were conceived:

- *RNA-disjoint*: all 173 proteins appear in every split (training, validation and testing datasets). RNA sequences are non-overlapping across splits.
- *Protein-disjoint*: proteins are non-overlapping across splits. RNA sequences may appear across splits due negative generation strategy.

Description of eCLIP data is presented in the respective tables below.

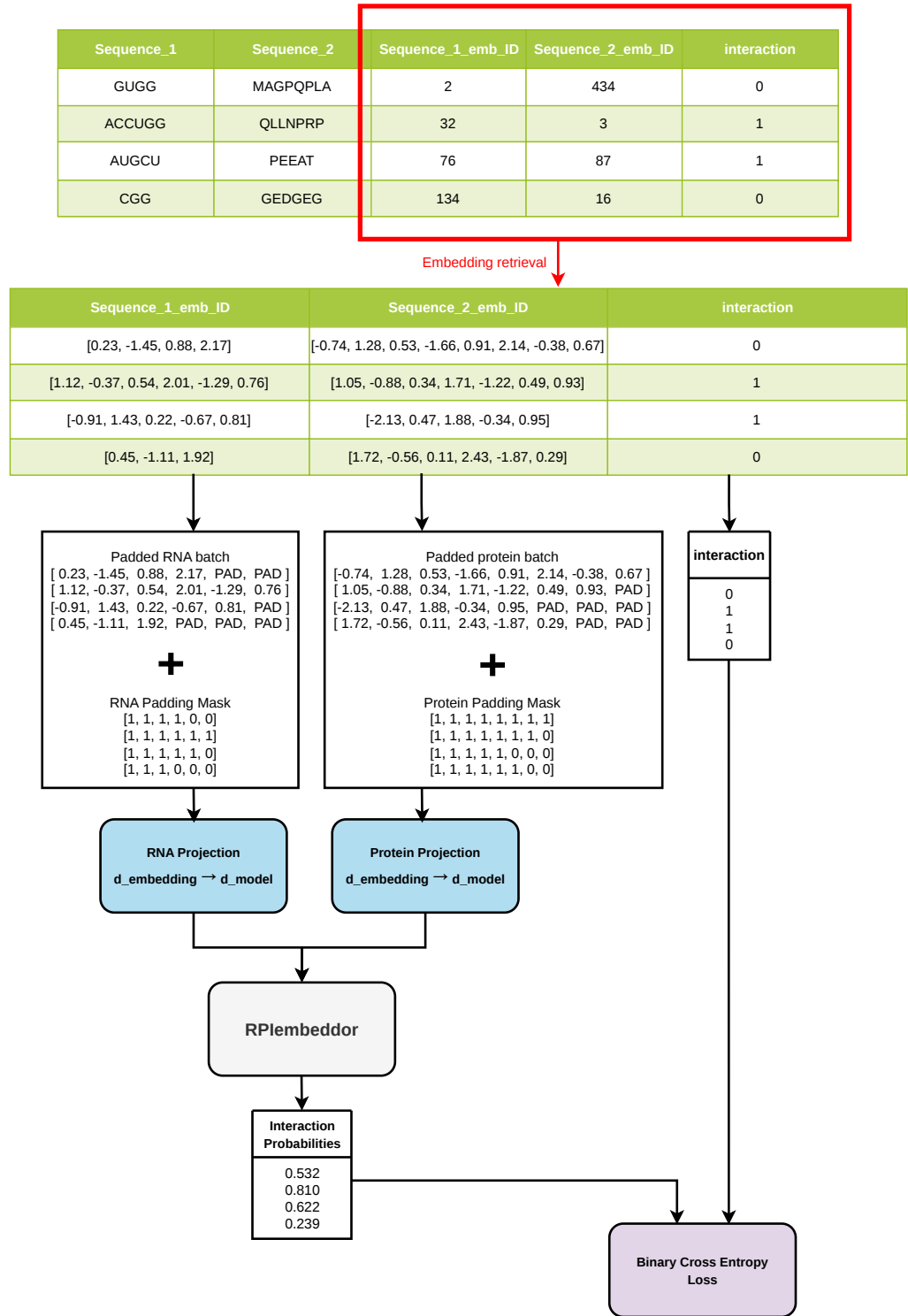
Table 4: Summary statistics of the eCLIP dataset (fixed length - 151 nts).

Property	Value
<i>General</i>	
Total RNA–protein pairs	1,472,694
Positive interactions	736,347 (50.0%)
Negative interactions	736,347 (50.0%)
<i>RNA sequences</i>	
Unique RNA sequences	708,729
RNA length	151 nt (fixed)
Mean pairwise identity	25.24%
Mean pairwise Hamming distance	112.9 / 151
Identical pairs (sampled)	0.000%
<i>Protein sequences</i>	
Unique protein sequences	173
Min length	126 aa
Max length	2,804 aa
Mean length	634.3 aa
Median length	577.0 aa
Std length	380.0 aa
<i>Pairs per protein</i>	
Min	148
Max	52,920
Mean	8,512.7
Median	4,090.0
Std	10,577.5

Table 5: Dataset split statistics for the two splitting strategies.

Property	RNA-disjoint split			Protein-disjoint split		
	Train	Val	Test	Train	Val	Test
Total pairs	1,030,828	220,981	220,885	1,072,274	192,412	208,008
Positive pairs	515,380	110,441	110,526	536,137	96,206	104,004
Negative pairs	515,448	110,540	110,359	536,137	96,206	104,004
% of dataset	70.0%	15.0%	15.0%	72.8%	13.1%	14.1%
Unique RNAs	496,110	106,309	106,310	598,640	169,778	182,781
Unique proteins	173	173	173	121	25	27

For the model, the actual inputs from both datasets are the same, only embeddings from the sequences in the batch, instead of raw sequences. Figure below is an overview on how inputs from a dataset are fed into the forward pass of RPIembeddor. Unique sequence identifiers are mapped to RNA and protein embeddings, which are dynamically padded within each batch and accompanied by padding masks. "Linear reduce" projection layers map embeddings into the Transformer latent space before processing by RPIembeddor. Model predictions are optimized using binary cross-entropy loss.



28 **Figure 4:** Inputs from datasets used by the forward pass of the model.

4.4 Original RNAInterAct dataset generation

We were provided with the github private repository containing the project code of RPIembeddor, the code to generate the curated RPI dataset used in the paper and also the last obtained checkpoint from the paper.

Unfortunately, this checkpoint could not be used to simply reproduce the results by doing inference, since the corresponding embeddings were missing.

Therefore, we had to regenerate the curated dataset from scratch from the genomic databases using python scripts and jupyter notebooks present for this in the project code. Our followed steps to achieve this are presented in the figure below.

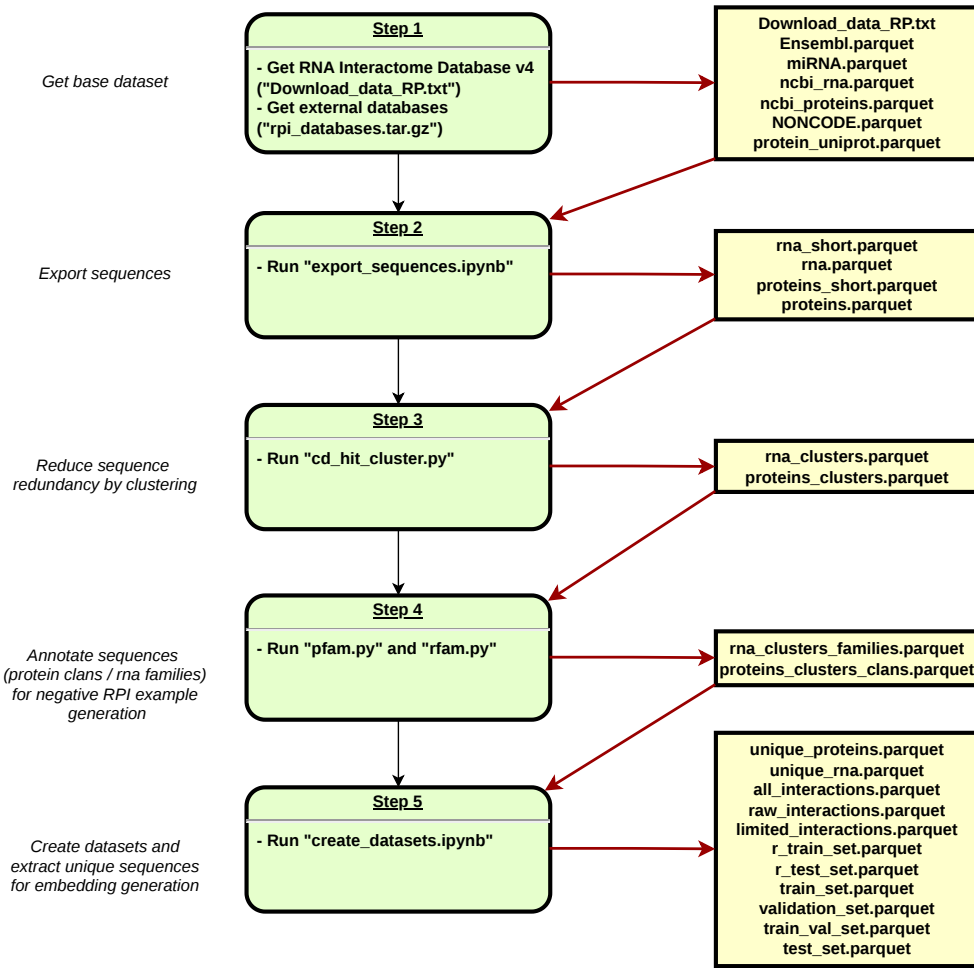


Figure 5: Our data pipeline for paper's curated dataset generation.

In the repository containing the code of the RPIembeddor were still present the original notebooks to generate the dataset with their output logs. Some output logs of these notebooks contained some counts and numbers related to statistics from the code running. Therefore, as we executed the data generation pipeline we compared these output logs in the jupyter notebooks to see if the the numbers match. Numbers were very similar. This suggested that the data was almost identical to the original data used in the paper.

Also, we installed the versions of the libraries, needed for the data generation, that were mentioned in the original paper trying to ensure maximal reproducibility.

4.5 Baseline model training and testing

Once we had the relevant dataset files and correct embeddings, we were able to train with the hyperparameters reported on the paper. The training process was regarded correct as per the shape of learning curves as the figure below shows:

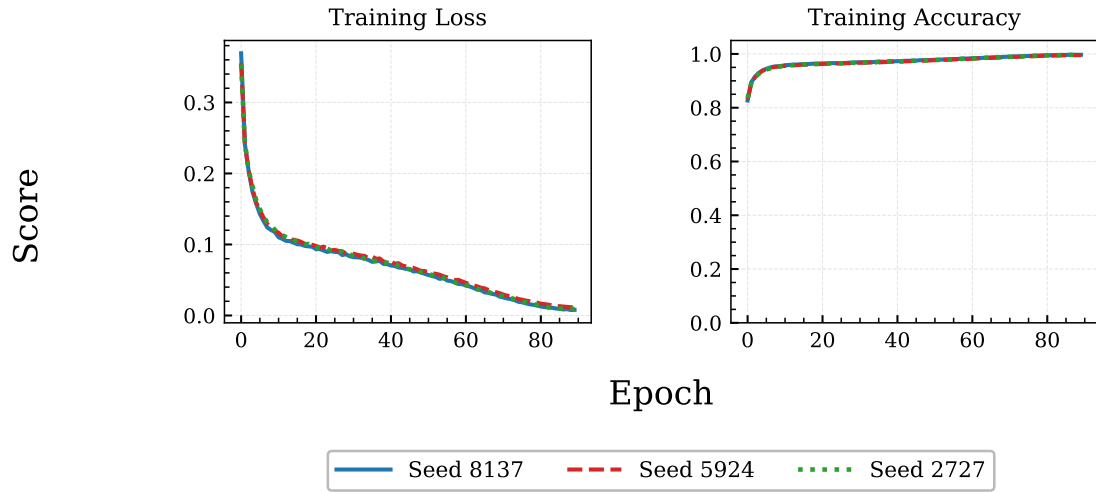


Figure 6: Learning curves across three random initializations.

Test performance is shown in Results and Discussion chapter.

Unexpectedly high test performance inspired us to run simpler model baselines to have a reference.

4.5.1 Logistic Regression Baseline to compare to the RPIembeddor baseline

As a simple baseline, a logistic regression classifier was trained on the same data as the baseline. For each RNA-protein pair, token-level embeddings were mean pooled across the sequence dimension to obtain fixed-length representations. The resulting RNA and protein vectors were then concatenated into a joint feature representation. A logistic regression model with ℓ_2 regularization was trained using the implementation provided by scikit-learn [41]. The model outputs an interaction probability for each RNA-protein pair, which was converted into a binary prediction using a threshold of 0.5.

Performance was evaluated using the same metrics as the neural network models, namely AUROC, AUPR, accuracy, F1-score, precision, and recall. Despite its simplicity, this baseline provides a useful reference point for assessing how much predictive information is already contained in the pretrained embeddings without explicitly modeling RNA-protein interactions.

4.5.2 Multi-Layer Perceptron Baseline on Frozen Foundation Model Embeddings

To provide a stronger non-linear baseline, a multi-layer perceptron (MLP) classifier was trained on frozen RNA and protein foundation model embeddings. Token-level embeddings were mean pooled across the sequence dimension to obtain fixed-length sequence representations. The resulting RNA and protein vectors were concatenated into a joint representation of each RNA-protein pair.

The classifier consisted of two fully connected hidden layers with 512 and 256 neurons, respectively. Each hidden layer was followed by batch normalization, ReLU activation, and dropout regularization with a dropout rate of 0.3. A final linear output layer produced a single logit corresponding to the predicted interaction probability.

Model parameters were optimized using the Adam optimizer with a learning rate of 10^{-3} , weight decay of 10^{-4} , and binary cross-entropy loss with logits. The model was trained for 90 epochs, and the checkpoint achieving the lowest validation loss was selected for final evaluation. Compared with the logistic regression baseline, the MLP is capable of modeling higher-order non-linear relationships between RNA and protein embedding features while remaining considerably simpler than the transformer-based RPIembeddor architecture.

4.5.3 Convolutional Neural Network Baseline on Frozen Foundation Model Embeddings to compare to the RPIembeddor baseline

To provide another baseline between the linear logistic regression model and the proposed transformer-based architecture, a convolutional neural network (CNN) classifier was trained on frozen RNA and protein foundation model embeddings. As in the logistic regression baseline, sequence-level representations were obtained by mean pooling the token embeddings across the sequence dimension, resulting in a fixed-dimensional vector for each RNA and protein sequence. The pooled RNA and protein embeddings were concatenated to form a joint representation of the RNA–protein pair. Unlike the logistic regression baseline, which models only linear relationships between the input features and the interaction label, the CNN baseline introduces non-linear feature extraction through stacked convolutional layers. The input feature vector was treated as a one-dimensional signal and processed using two one-dimensional convolutional layers with batch normalization and ReLU activation functions. A global max-pooling operation was subsequently applied to aggregate the most informative activation patterns across the feature space. The pooled representation was passed through a fully connected hidden layer with dropout regularization, followed by a single output neuron producing the interaction probability. Model parameters were optimized using the Adam optimizer with a learning rate of 10^{-3} and binary cross-entropy loss. The CNN was trained to 90 epochs as the baseline. The figure below shows the CNN architecture.

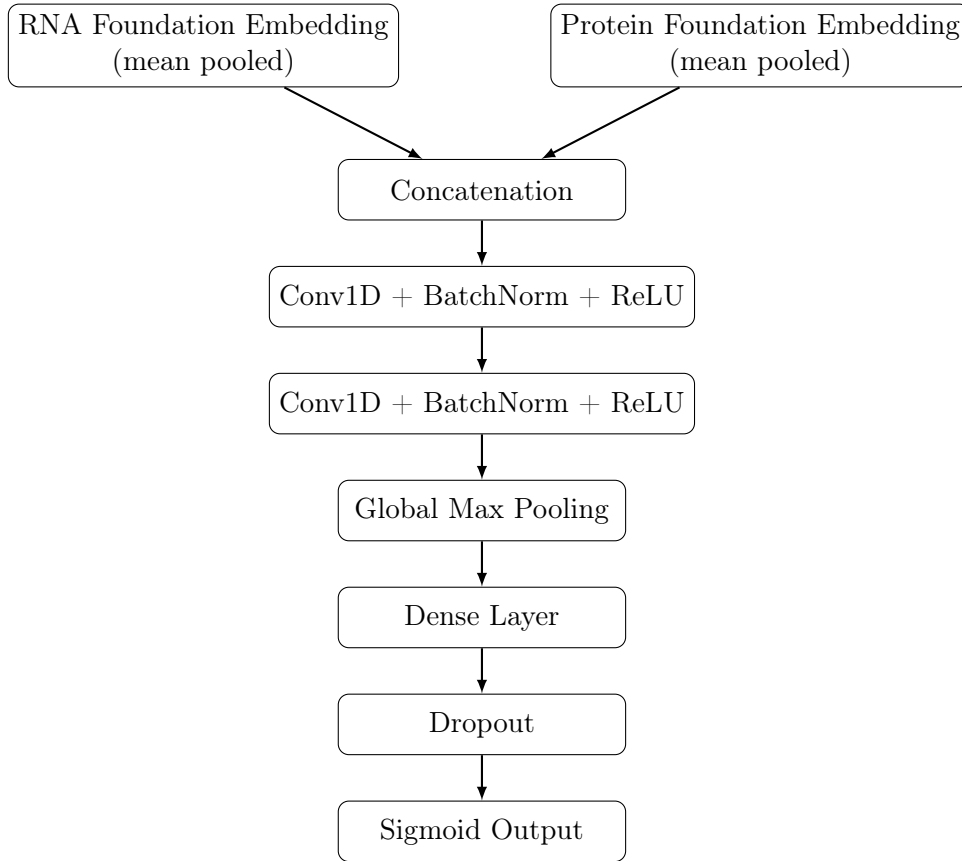


Figure 7: CNN baseline architecture operating on frozen RNA and protein foundation model embeddings.

4.5.4 Hyperparameter Optimization on baseline

In the original RPIembeddor code, there was a dedicated script for Hyperparameter Optimization (HPO) using the NePS framework [42], employing the Hyperband algorithm [43] to allocate training budgets adaptively across candidate configurations.

Since the baseline training worked, we decided to run this HPO script with the default paper’s hyperparameters. Our intention with this was to observe if HPO could yield better performance on the baseline.

The best found configuration is shown at table below:

Table 6: Best found hyperparameter configuration.

Hyperparameter	Value
Learning rate	0.00039457092606005325
Weight decay	0.0005081310266379466
Dropout	0.16244020564524297
Epochs	10

We trained again the baseline with this found hyperparameters and test performance was improved slightly. This is shown in Results and Discussion Chapter.

The parameters for the HPO experiment are in the following tables:

Table 7: Hyperparameter search space used for NePS Hyperband optimization.

Hyperparameter	Range
Learning rate	$[10^{-5}, 10^{-2}]$
Weight decay	$[10^{-5}, 10^{-2}]$
Dropout	$[0, 1]$
Epochs (fidelity)	$[3, 90]$

Table 8: Fixed model and training parameters during hyperparameter optimization.

Parameter	Value
Embedding dimension (d_{model})	256
Number of attention heads	2
Feed-forward dimension	20
Number of encoder layers	1
Batch size	8
Warmup steps	1000
Maximum scheduler period (T_{max})	90
Random seed	150
GPUs	2

4.6 Handling sequences of longer than 1022 residues

As we introduced more diverse data to train the RPIembeddor (eCLIP), having longer than 1022 aminoacids protein sequences was possible, unlike the RNAInterAct dataset where such long sequences were intentionally filtered from the dataset generation pipeline.

As presented in the Background chapter, normally, the foundation models have maximum lengths for the sequences to process as inputs (context length). It is highly encouraged not to exceed these limits, as undesired outcomes in the model expected processing behavior occur, such as trimming the sequence to force the maximum length or obtaining low quality embeddings since the training distribution had such limit.

As a result, we took inspiration of an approach in the literature: the sliding window routine used by Brandes et al. [44]. In this approach they do a sliding window to cover protein sequences longer than 1022 aminoacids by subdividing longer sequences into overlapping 1,022 amino acid windows with at least 511 amino acid overlap.

More specifically according to their description: " Each protein sequence was tiled by iteratively generating 1,022 amino acid window from both ends of the sequence such that consecutive windows had exactly 511 amino acid overlap until windows from both ends met at the center. If the overlap between the central windows was less than 511 amino acids, an additional 1,022 amino acid window was added at the center. ".

The reason on this was the following: A residue near the edge of a window gets a poor quality embedding since the attention mechanism has little context on one side. By using overlapping windows and averaging the scores for edge residues, it is ensured that every residue is represented in a context where it is closer to the center of at least one window, where the attention context is richest. The overlap was specifically designed to mitigate these edge effects.

A residue at position 1 of a window only has context from one side, so its embedding is of lower quality than a residue at position 511 which has 511 residues of context on both sides. With non-overlapping windows, every boundary residue suffers from this. With overlapping windows, boundary residues get embedded twice (once near the edge of one window and once more toward the center of the next). Averaging those two embeddings gives a better representation than either edge embedding alone. Non-overlapping windows would leave every boundary residue with a poor one-sided embedding.

On the figure below, there is a representation of the weighted sliding window routine to embed long sequences.

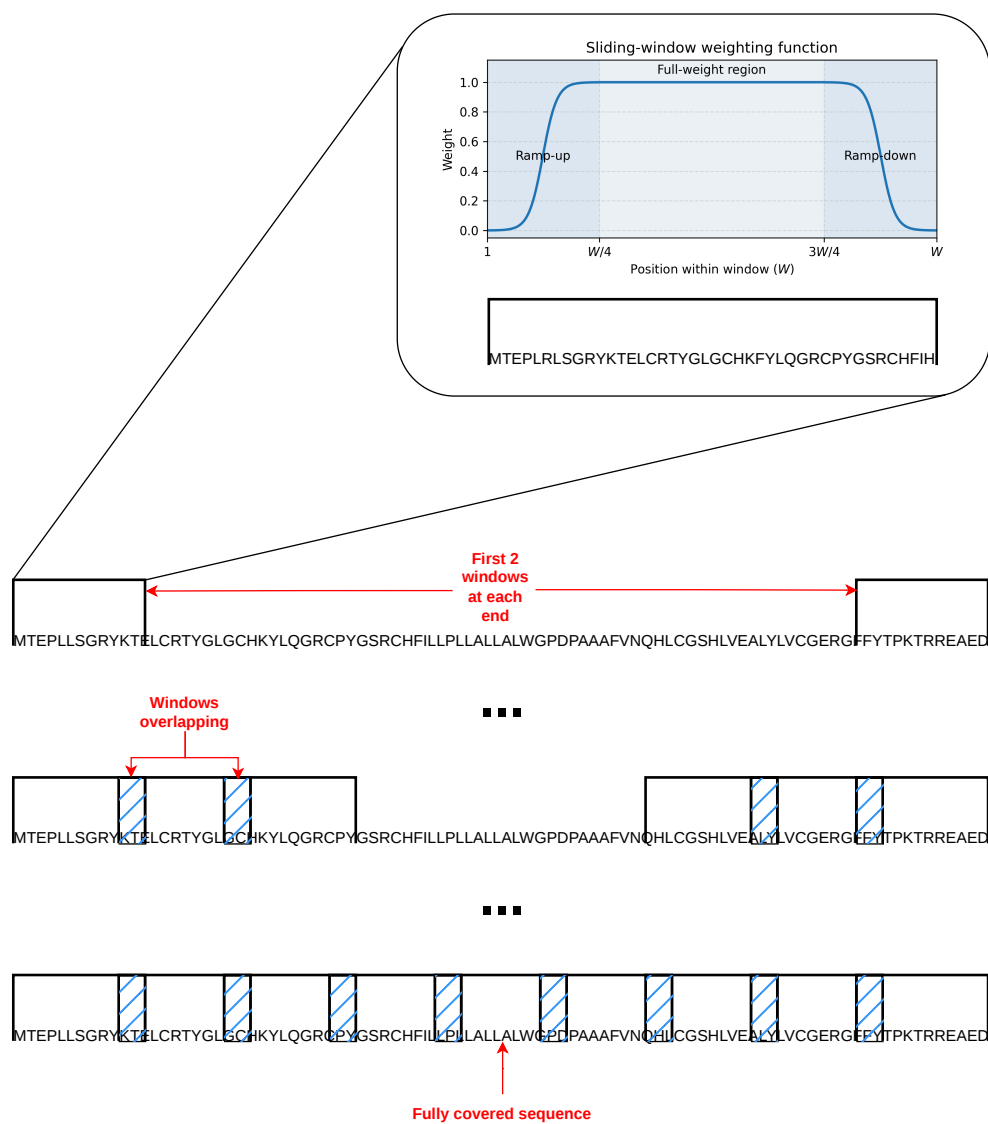


Figure 8: Sliding window routine.

4.7 Training with different frozen embeddings combinations

Since the publication of the RPIembeddor, new foundation models claiming to perform at a State of The Art level were published. Therefore, observing the effect of those on the RPIembeddor was considered insightful and interesting.

We decided to use the hyperparameters of the baseline (the ones used in the RPIembeddor paper), since they resembled standard working hyperparameters for deep learning. We trained to 90 epochs with early stopping (patience 10) for every combination of foundation models.

Results are shown in the following chapter for RNAInterAct and eCLIP data.

4.8 Ablation

We decided to test the assumption of the foundation models having truly general genomic knowledge. We decided to observe this by substituting the embeddings with one hot encoded representations of the sequences.

5 Results and discussion

Initially, we started investigating with the original data from RPIembeddor: RNAInterAct.

Test performances were obtained across three random initializations. Values are reported as mean $\pm$ standard deviation.

5.1 RNAInterAct data

Before evaluating the effect of different foundation model embeddings with RPIembeddor, it is important to note that different splitting strategies impose different generalization requirements and can substantially influence the performance of machine learning models. For RNAInterAct, we considered two settings: the original RNA family-disjoint split, where RNA families are strictly separated between training, validation, and test sets, and, as we present later in this discussion, a new, relaxed split in which training and validation samples were randomly assigned while preserving the original RNA family-disjoint test set. The first setting represents a stricter out-of-distribution generalization scenario, whereas the second reduces the distribution shift between training and validation data.

5.1.1 Simpler baselines results

First, we considered evaluating simple baseline models on RNAInterAct data to obtain a reference estimate of task difficulty before introducing the more complex RPIembeddor architecture and different foundation model embeddings. The results of these simple models can be seen below.

Table 9: Performance of simple baseline models on the RNAInterAct RNA family-disjoint split as reference.

Metric	CNN	LogReg	MLP
AUROC	0.826 ± 0.092	0.842 ± 0.000	0.943 ± 0.006
AUPR	0.762 ± 0.118	0.777 ± 0.000	0.921 ± 0.006
Accuracy	0.743 ± 0.102	0.767 ± 0.000	0.872 ± 0.010
F1 Score	0.496 ± 0.375	0.696 ± 0.000	0.833 ± 0.016
Precision	0.839 ± 0.064	0.699 ± 0.000	0.834 ± 0.014
Recall	0.437 ± 0.353	0.694 ± 0.000	0.833 ± 0.035

Surprisingly, even well established baselines with less capacity than the RPIembeddor performed decently. One reason for this is because the RNAInterAct data is unbalanced. This gives signal to explode for all simpler models. However, the MLP remarkably stands out. In this comparison plays a crucial role which model can handle better concatenated, mean pooled representations of RNA and protein embeddings. The MLP seems to be better fitted for receiving this kind of features. Logistic regression could arguably somewhat struggle, since the relations between the data is probably not linear, totally exceeding the logistic regression capacity. Although CNN can also handle non linear relationships on the data, due to the nature of RPI interaction, which normally involves local motifs, it is hypothesized that, at the mean pooling step on the embeddings, the kind of signal that a CNN could explode is diminished and as a result CNN can also struggle. Now, having a reference on our task with simpler models we could proceed to observe the baseline performance.

5.1.2 RPIembeddor baseline test results and HPO

The RNAInterAct experiments served as the first validation point for the experimental setup. The reproduced RPIembeddor baseline achieved strong test performance, indicating that the implementation and embedding loading pipeline were functional. Hyperparameter optimization (NePS with Hyperband) further improved the baseline, suggesting that the original configuration was not necessarily optimal for our training setup. However, the strong performance of simpler baselines, especially the MLP, indicates that high performance on this dataset does not necessarily require the full RPIembeddor architecture. This suggests that part of the predictive signal may already be accessible from the frozen embeddings without complex cross-attention modeling.

Results are shown in table below.

Table 10: Comparison between the default RPIembeddor configuration and the hyperparameter-optimized (HPO) RPIembeddor. Values are reported as mean $\pm$ standard deviation over 3 random initializations. Best performance for each metric is shown in bold.

Metric	RPIembeddor	RPIembeddor + HPO
AUROC	0.933 $\pm$ 0.006	0.949 $\pm$ 0.001
AUPR	0.864 $\pm$ 0.008	0.908 $\pm$ 0.003
Accuracy	0.876 $\pm$ 0.010	0.894 $\pm$ 0.002
F1 Score	0.843 $\pm$ 0.010	0.868 $\pm$ 0.003
Precision	0.824 $\pm$ 0.021	0.843 $\pm$ 0.003
Recall	0.864 $\pm$ 0.002	0.894 $\pm$ 0.004

The HPO yielded a smaller learning rate as well as less training epochs, among other hyperparameters. This could fight overfitting in the model training, leading to a better test performance. However, this optimization was only with respect to the RPIembeddor baseline setting and data. There was no guarantee that this had an equally positive effect with other kind of data as eCLIP. We only performed HPO on this setup.

Training RPIembeddor with RNAInterAct: Unstable learning curves

One important feature of the training of the RPIembeddor baseline, was that, as per the code and the paper, training and testing were only done with a training and testing set, respectively, totally leaving out a proper validation set. From this situation, following standard ML practices, we changed this and generated a new split from RNAInterAct: a validation dataset.

We trained again the baseline including this extra data split (validation dataset), and observed pronounced fluctuations (“zigzagging”) in validation loss and derived metrics such as F1-score, precision, and accuracy across epochs.

The RNAInterAct dataset employs an RNA family-wise split, ensuring that RNA families in the training, validation, and test sets are disjoint. This setup enforces a strict out-of-distribution (OOD) generalization scenario, where the model must extrapolate to unseen RNA families. We hypothesized that the observed instability in validation performance on RNAInterAct is primarily driven by this distribution shift rather than by deficiencies in the optimization process. Under OOD conditions, small parameter updates during training can lead to disproportionately large changes in validation performance, as the model alternately captures or fails to capture features that generalize across RNA families. This effect is further amplified by the class

imbalance present in RNAInterAct (approximately 3:1 negative-to-positive ratio), which increases the sensitivity of metrics such as F1-score and precision to minor prediction changes.

To further assess this hypothesis, an ablation experiment was conducted in which the strict family-wise separation was relaxed by mixing training and validation data (random split only between training and validation dataset, test set remained the same, with RNA families disjointness with respect to training and validation sets). Under this setting, validation curves exhibited significantly reduced variance, supporting the conclusion that the instability is largely data-driven rather than purely a consequence of hyperparameter selection.

Finally, early stopping (patience 10) was introduced to mitigate overfitting effects observed in later training stages. Fixed-length training (e.g., 90 epochs) was found to be suboptimal, as validation performance often degraded after a certain point despite continued improvements in training loss.

The observed validation instability on RNAInterAct, presented in figure below, should not be interpreted solely as a training deficiency. Instead, it reflects the inherent difficulty of the task under strict distributional shifts.

The validation behavior on RNAInterAct revealed that the difficulty of the task is strongly affected by the splitting strategy. Under the RNA family-disjoint split, validation curves showed pronounced fluctuations, whereas a relaxed random train-validation split produced more stable training dynamics. This supports the interpretation that the instability was largely data-driven rather than purely caused by optimization problems. The family-disjoint split represents a stricter out-of-distribution setting, where the model must generalize to RNA families not observed during training. Therefore, unstable validation performance should not be interpreted only as a training failure, but also as a consequence of the demanding generalization regime.

We tested the best (having the least validation loss) and last trained checkpoint from each dataset split strategy, expecting a better performance for the training with more stable learning curves. Test performance is shown in table below.

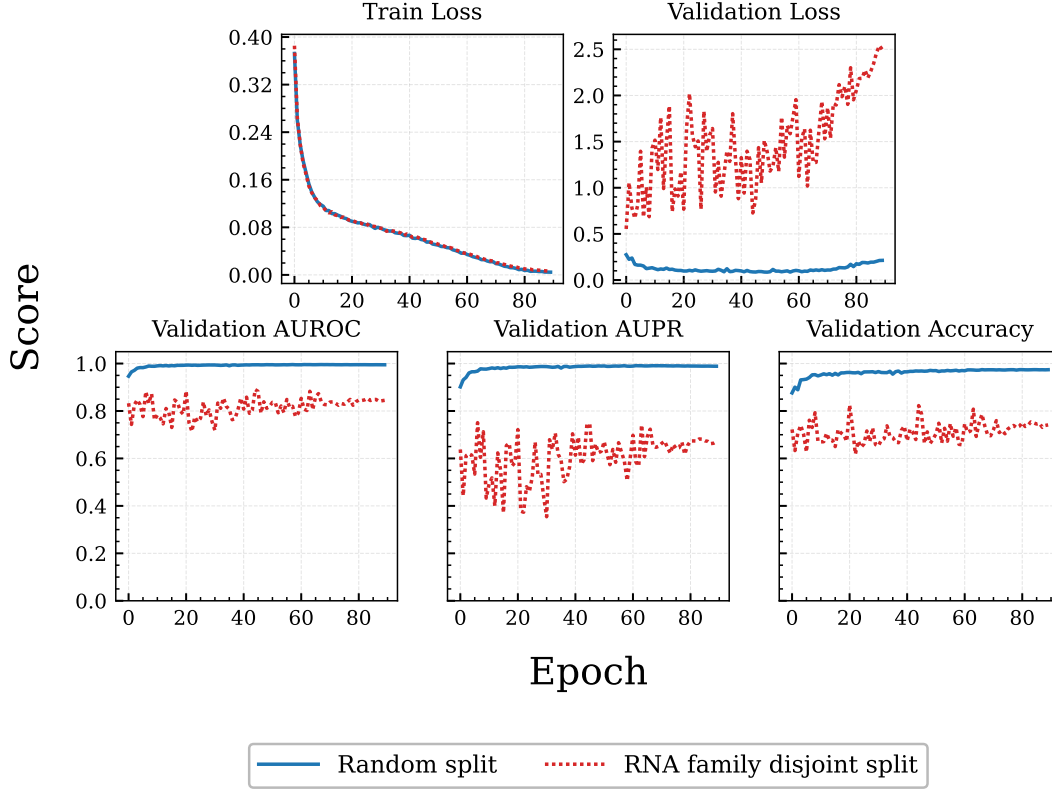


Figure 9: Unstable behavior in baseline validation training metrics with original dataset split compared to stable behavior when using an easier dataset split at baseline training.

Table 11: Performance comparison between RNA Family Disjoint and Random splitting strategies.

Metric	RNA Family Disjoint		Random	
	Best	Last	Best	Last
AUROC	0.942	0.939	0.938	0.935
AUPR	0.916	0.883	0.905	0.877
Accuracy	0.868	0.885	0.873	0.878
F1	0.830	0.853	0.836	0.842
Precision	0.827	0.843	0.831	0.840
Recall	0.832	0.863	0.840	0.844

Test results were unexpectedly similar, suggesting unstable training curves could not affect so negatively training learning. This unstable curves at training seems not to be an actual problem, probably because the hyperparameters prevent the training

from completely diverging, still allowing some instability.

We kept this additional RNAInterAct data split for subsequent evaluations.

5.1.3 Comparing embedding combinations on each split

We proceeded with the evaluations on every embedding combination with the RPIembeddor. The test performances of RPIembeddor on RNAInterAct data with the original split and our proposed easier split is shown below.

Table 12: Comparison of protein and RNA foundation model pairings on RNAInterAct. Results are shown for the RNA family-disjoint and random splits.

Protein FM	RNA FM	RNA Family-Disjoint		Random Split	
		AUPR	AUROC	AUPR	AUROC
ESM-2	LAMAR	0.871±0.016	0.917±0.008	0.831±0.009	0.904±0.004
ESM-2	RNA-FM	0.921±0.013	0.943±0.009	0.901±0.013	0.938±0.004
ESM-2	RNAElectra	0.928±0.001	0.948±0.002	0.930±0.007	0.951±0.003
ESM-C	LAMAR	0.539±0.079	0.667±0.060	0.684±0.012	0.779±0.023
ESM-C	RNA-FM	0.645±0.043	0.737±0.038	0.754±0.027	0.832±0.032
ESM-C	RNAElectra	0.582±0.053	0.685±0.037	0.737±0.005	0.807±0.008
LC-PLM	LAMAR	0.727±0.040	0.828±0.036	0.809±0.009	0.889±0.003
LC-PLM	RNA-FM	0.848±0.052	0.902±0.030	0.874±0.014	0.926±0.006
LC-PLM	RNAElectra	0.854±0.021	0.902±0.017	0.902±0.006	0.942±0.003
V-ESM	LAMAR	0.853±0.016	0.909±0.009	0.817±0.003	0.895±0.002
V-ESM	RNA-FM	0.887±0.016	0.919±0.009	0.898±0.005	0.938±0.003
V-ESM	RNAElectra	0.898±0.006	0.928±0.006	0.914±0.007	0.944±0.003

The results on RNAInterAct reveal a clear trend on the protein side. Across both splitting strategies, ESM-2 consistently produced the strongest results and appeared substantially more robust than the alternative protein encoders. In contrast, ESM-C showed a marked degradation in performance, while LC-PLM and V-ESM generally achieved intermediate results. This suggests that the representations learned by ESM-2 are particularly well aligned with the interaction patterns present in RNAInterAct and can be effectively exploited by the RPIembeddor architecture.

On the RNA side, RNAElectra consistently achieved the best performance when paired with strong protein encoders. In particular, the ESM-2 + RNAElectra combination obtained the highest AUROC and AUPR values under both the RNA family-disjoint and random split settings. RNA-FM remained highly competitive, whereas LAMAR generally underperformed on this dataset despite showing promising results in other settings. These observations indicate that the optimal RNA encoder is

dataset-dependent and suggest that RNAElectra captures sequence characteristics that are especially informative for the interaction patterns represented in RNAInterAct.

To complement our evaluation on the downstream training with RPIembeddor, we considered the unsupervised metrics to see if intrinsic quality on the embeddings seemed to correlate with downstream performance. Results are shown in tables below.

Table 13: Unsupervised representation quality metrics for protein foundation models on the RNAInterAct dataset.

Protein FM	Hidden Dim.	IsoScore	RankMe	NESum	StableRank
ESM-2	640	0.057	113.371	4.176	1.088
ESM-C	1152	0.003	14.803	1.458	1.168
LC-PLM	1536	0.011	4.747	2.995	1.008
V-ESM	640	0.031	108.069	2.930	1.098

Table 14: Unsupervised representation quality metrics for RNA foundation models on the RNAInterAct dataset.

RNA FM	Hidden Dim.	IsoScore	RankMe	NESum	StableRank
LAMAR	768	0.029	136.757	1.967	1.131
RNA-FM	640	0.038	97.855	3.011	1.042
RNAElectra	512	0.001	12.919	1.142	1.186

The unsupervised embedding metrics provided useful diagnostics of the embedding spaces, but they did not fully predict downstream RPIembeddor performance. For example, RNAElectra showed relatively low intrinsic embedding-quality scores but performed strongly in several RNAInterAct experiments. Conversely, LAMAR showed strong intrinsic metrics and also performed well on eCLIP. These results indicate that metrics such as IsoScore, RankMe, NESum, and StableRank capture important geometric properties of the embeddings, but they are not sufficient on their own to determine whether an embedding will be useful for a specific downstream interaction prediction task.

One interesting observation is that RNAElectra, despite having the lowest unsupervised metrics across the board, was often the best-performing RNA encoder for the RNAInterAct data setup. These metrics suggest that high intrinsic embedding diversity does not necessarily translate into better downstream interaction prediction. Also LAMAR appeared to have good intrinsic embedding quality, and this was congruent with the downstream evaluation presented above.

5.2 eCLIP data

The RNAInterAct experiments motivated the introduction of eCLIP data as a second evaluation setting. Compared with RNAInterAct, the eCLIP dataset provides a much larger number of RNA sequences and is directly derived from experimental binding assays. This changed the nature of the prediction problem: instead of evaluating only on a relatively small curated interaction dataset, the model was tested in a setting with substantially higher RNA diversity and protein-specific binding profiles.

5.2.1 Observing performance of a one hot encoded CNN using only RNA sequences

Given the abundant RNA sequences from this new dataset. We questioned whether such a diverse corpus of RNA context could on its own serve to solve another related task: Check whether a given RNA sequence seems to be a protein binder or non binder.

For this, we took inspiration on a CNN using one hot encoded RNA sequences present in [33], where they introduced this model in order to work with CLIP data. It is important to note that this CNN is **not** the same as the one presented in Methods chapter.

Specifically, A lightweight convolutional neural network (CNN) baseline was trained using only RNA sequence information to provide a protein-specific reference model.

RNA sequences of length 151 nucleotides were converted into one-hot encoded representations with four channels corresponding to A, C, G, and U/T nucleotides. Unknown nucleotides were encoded as zeros. The resulting input tensor had dimensions 4×151 .

The CNN architecture consisted of three one-dimensional convolutional blocks with 64, 96, and 128 filters, respectively. Each convolutional layer was followed by batch normalization, ReLU activation, max pooling, and dropout regularization. The convolutional layers were followed by a fully connected layer of 256 units and a final linear output neuron.

Models were trained using the Adam optimizer with learning rate 0.001 and binary cross-entropy loss with logits (BCEWithLogitsLoss). Training was performed for up to 90 epochs with batch size 256. Learning rate scheduling was used. Early stopping with a patience of 10 epochs was applied to prevent overfitting.

The best model was selected according to validation AUROC and subsequently evaluated on the held-out test set. Final performance was reported using AUROC, AUPR, accuracy, and F1 score.

We used this CNN for our new task (only using RNA sequences), but with a subset of the first 8 proteins of the eCLIP data with **random** negatives. The results are shown below.

Table 15: Performance of the CNN baseline using one-hot encoded RNA sequences.

Model	AUROC	AUPR	F1	Accuracy
CNN_Baseline_OneHot	0.906	0.904	0.827	0.825

The high performance of the RNA-only CNN baseline on the initial eCLIP setup suggested that the task may have been easier than intended. Since the model only received RNA sequence information, strong performance indicated that it could separate positive RNA regions from randomly generated negative regions without using protein information. This raised the possibility that the model was learning general sequence properties of binding-like regions rather than protein-specific RNA-protein interaction patterns.

Since we had initially randomly generated the negatives on our eCLIP data, we interpreted that this CNN exploded this setting in the following way: The model then had to learn solely whether a given RNA sequence looked different than random noise from the sequence level to determine whether a given RNA could potentially be an interacting sequence or not. Such a simple heuristic could facilitate the difficulty of this task and enable simple models to excel on it with ease. Therefore, we decided to change our negative generation strategy.

Negative generation: Taking other positives as negatives

Instead of sampling random negative regions from gene regions occupied by positives (previous negatives), the new negatives were decided to be actual positives, but from other proteins. This heuristic had the intention to force the model learn actual RPI specificity. Now, the model would not only have to differentiate a plausible interacting sequence from random noise but actually to recognize conformations of interacting sequences for each protein, i.e. a given RNA sequence has the specific structure to be a binder for protein A, but not for protein B.

Such new heuristic was more biologically meaningful than the random previous approach. We trained again this CNN baseline on our eCLIP data with new negatives. In this new training, a separate CNN was trained for each eCLIP protein dataset corresponding to 173 in total. The following table shows performance of the RNA-only CNN baseline evaluated across 173 protein-specific datasets. Each model was

trained using one-hot encoded RNA sequences only, without any protein information. Reported values summarize the distribution of test performance across proteins.

Table 16: RNA one-hot encoded baseline performance across 173 single protein datasets from the eCLIP data with new negatives.

Metric	Mean	Std	Median	Min	Max
Accuracy	0.762	0.126	0.793	0.500	0.975
AUROC	0.873	0.091	0.895	0.596	0.997
AUPR	0.867	0.096	0.888	0.545	0.997
F1 Score	0.702	0.225	0.780	0.000	0.975

With these results, we could confirm that the new negatives made the data more difficult to learn from. We observed that the performances of these CNN’s were a bit spread. We attributed this to the dataset size on which each CNN was trained on. Some proteins have 40,000 RNA sequences associated while other less than 1000. This evaluation also implies that this simpler task, not involving protein representations, is solvable with the correct model without having to use complex attention layers. From here, we continued examining different embeddings for RPIembedder with the improved eCLIP data (new negatives). Our intention was now to move to the setting having many proteins at the same time, such that generalization across proteins could be tested.

5.2.2 Benchmarking foundation model combinations

Before comparing foundation model embeddings on eCLIP, it is important to establish the relative difficulty of the two splitting strategies considered in this work. In the RNA-disjoint split, all proteins are present in the training, validation, and test sets, while RNA sequences are non-overlapping across splits. This setting evaluates the ability of a model to generalize to unseen RNA molecules while preserving exposure to the same set of proteins during training. In contrast, the protein-disjoint split represents a substantially more challenging scenario, as proteins are strictly separated between training, validation, and test sets. Under this setting, the model must generalize to entirely unseen proteins and cannot rely on protein-specific interaction patterns observed during training.

Similarly with the RNAInterAct data, we first tried the same 3 simpler baselines to observe how they performed on the eCLIP data. The results are shown in table below.

Table 17: Performance of baseline models across the eCLIP data splits. Abbreviations: Prot. Disj. = protein-disjoint, RNA Disj. = RNA disjoint.

Split	Model	AUROC	AUPR	F1	Accuracy
eCLIP Prot. Disj.	CNN	0.589±0.005	0.566±0.005	0.591±0.017	0.567±0.006
	LogReg	0.606±0.000	0.573±0.000	0.515±0.000	0.559±0.000
	MLP	0.703±0.006	0.664±0.004	0.578±0.014	0.630±0.007
eCLIP RNA Disj.	CNN	0.820±0.002	0.816±0.002	0.750±0.002	0.735±0.001
	LogReg	0.643±0.000	0.594±0.000	0.598±0.000	0.599±0.000
	MLP	0.912±0.000	0.909±0.000	0.831±0.001	0.828±0.001

The comparison with simpler baselines across the different splits highlights that dataset design strongly determines task difficulty. In the RNA disjoint eCLIP split and the RNAInterAct random split, simple models achieved high performance, suggesting that these settings contain signals that can be exploited without a complex interaction architecture. In contrast, the eCLIP protein-disjoint split produced much lower baseline performance, indicating that generalization to unseen proteins is substantially harder. These results show that simpler baselines are not merely references, but important diagnostic tools for identifying whether a benchmark requires genuine interaction modeling.

From this again, we proceeded to the evaluations on embeddings combinations with the RPIembeddor now on eCLIP data. Results are shown below.

Table 18: Performances across protein and RNA foundation model combinations with eCLIP protein disjoint split.

Protein FM	RNA FM	AUROC	AUPR	F1	Accuracy
ESM-2	LAMAR	0.720±0.020	0.691±0.019	0.625±0.042	0.649±0.022
ESM-2	RNA-FM	0.696±0.006	0.670±0.011	0.547±0.008	0.615±0.006
ESM-2	RNAElectra	0.654±0.018	0.618±0.015	0.636±0.030	0.612±0.020
ESM-C	LAMAR	0.679±0.024	0.641±0.024	0.447±0.121	0.585±0.037
ESM-C	RNA-FM	0.668±0.022	0.626±0.019	0.542±0.024	0.599±0.007
ESM-C	RNAElectra	0.695±0.012	0.658±0.015	0.602±0.019	0.630±0.013
LC-PLM	LAMAR	0.642±0.024	0.608±0.016	0.542±0.088	0.593±0.027
LC-PLM	RNA-FM	0.604±0.031	0.574±0.035	0.504±0.025	0.563±0.028
LC-PLM	RNAElectra	0.614±0.001	0.575±0.005	0.608±0.052	0.581±0.006
V-ESM	LAMAR	0.674±0.007	0.625±0.007	0.493±0.015	0.587±0.005
V-ESM	RNA-FM	0.666±0.010	0.620±0.014	0.514±0.043	0.591±0.009
V-ESM	RNAElectra	0.626±0.007	0.581±0.012	0.587±0.031	0.595±0.003

Table 19: Performance of protein and RNA foundation model combinations on the eCLIP RNA disjoint split.

Protein FM	RNA FM	AUROC	AUPR	F1	Accuracy
ESM-2	LAMAR	0.937±0.009	0.933±0.010	0.862±0.013	0.861±0.012
ESM-2	RNA-FM	0.923±0.012	0.919±0.013	0.846±0.014	0.844±0.013
ESM-2	RNAElectra	0.907±0.013	0.901±0.015	0.830±0.011	0.822±0.013
ESM-C	LAMAR	0.938±0.008	0.935±0.009	0.864±0.011	0.862±0.010
ESM-C	RNA-FM	0.923±0.011	0.919±0.012	0.847±0.012	0.842±0.013
ESM-C	RNAElectra	0.900±0.014	0.895±0.015	0.819±0.016	0.815±0.016
LC-PLM	LAMAR	0.947±0.000	0.945±0.000	0.876±0.000	0.874±0.000
LC-PLM	RNA-FM	0.920±0.019	0.917±0.020	0.843±0.023	0.839±0.023
LC-PLM	RNAElectra	0.886±0.015	0.881±0.016	0.808±0.013	0.800±0.016
V-ESM	LAMAR	0.944±0.007	0.942±0.007	0.873±0.009	0.870±0.009
V-ESM	RNA-FM	0.939±0.001	0.937±0.001	0.865±0.001	0.861±0.002
V-ESM	RNAElectra	0.892±0.010	0.886±0.011	0.815±0.010	0.806±0.011

The foundation model comparison shows that replacing the original RNA-FM and ESM-2 embeddings does affect downstream RPI prediction, but no single embedding model was universally superior across all datasets and splits. On the protein side, ESM-2 remained a consistently strong choice, especially in the more difficult settings. On the RNA side, the best-performing model depended on the dataset: RNAElectra performed strongly on RNAInterAct, whereas LAMAR was more competitive on eCLIP. This suggests that embedding usefulness is task- and data-dependent, and that benchmark performance from one biological setting does not necessarily transfer directly to another.

These results suggest that the embedding quality of ESM2 has decent signal for the protein side of RPI classification, since it appeared to be the best for our 4 different splits.

From the RNA side, there was no single RNA FM outperforming the other in all the data settings. RNAElectra performed better in the RNAInterAct data, whereas LAMAR did better on the more difficult eCLIP data.

We considered that the eCLIP evaluation was more robust, as this data is intrinsically harder to learn from due to the locality in their interaction motifs, as opposed to the heterogeneity from RNAInterAct.

As a complement to this comparison, we obtained unsupervised metrics for the embeddings for the eCLIP data too. Results are shown below.

Table 20: Unsupervised representation quality metrics for protein foundation models on the eCLIP dataset.

Protein FM	Hidden Dim.	IsoScore	RankMe	NESum	StableRank
ESM-2	640	0.056	41.829	4.000	1.083
ESM-C	1152	0.013	57.554	2.181	1.336
LC-PLM	1536	0.007	4.826	1.337	1.008
V-ESM	640	0.035	38.200	3.305	1.092

Table 21: Unsupervised representation quality metrics for RNA foundation models on the eCLIP dataset.

RNA FM	Hidden Dim.	IsoScore	RankMe	NESum	StableRank
LAMAR	768	0.047	196.197	3.934	1.135
RNA-FM	640	0.026	62.217	4.202	1.019
RNAElectra	512	0.003	6.478	1.569	1.012

Overall, the results suggest that improving RNA–protein interaction prediction is not simply a matter of replacing the original foundation models with newer alternatives. Performance depends strongly on the dataset, the negative sampling strategy, the split regime, and the downstream architecture. RNAInterAct was useful for reproducing and validating the original framework, but its interpretation was complicated by class imbalance, strong simple baselines, and split-dependent validation behavior. eCLIP provided a larger and more experimentally grounded setting, but required careful negative generation to avoid overly easy sequence-level shortcuts. Across both datasets, foundation-model embeddings influenced performance, but their usefulness was context-dependent rather than universal.

5.2.3 Ablation

Lastly, we performed the mentioned ablation in Methods chapter. Results are shown below.

Table 22: Performance of RPIembeddor trained using one-hot encoded protein and RNA representations on the eCLIP RNA disjoint split.

Metric	Score
AUROC	0.833
AUPR	0.824
Accuracy	0.750
F1 Score	0.754
Precision	0.743
Recall	0.766

The replacement of both RNA and protein foundation model embeddings with one-hot encoded representations resulted in a noticeable decrease in predictive performance. Although the model still achieved an AUROC of 0.833 and an AUPR of 0.824, all evaluation metrics were substantially lower than those obtained using pretrained foundation model embeddings.

This result suggests that a significant portion of the predictive power of RPIembeddor originates from the information captured by the pretrained sequence representations. Foundation models encode evolutionary, structural, and contextual sequence features that are not directly available in one-hot encodings, where each nucleotide or amino acid is represented independently, without any notion of similarity or biological context.

Nevertheless, the fact that the model remains capable of achieving reasonable performance using only one-hot representations indicates that the architecture itself contributes meaningfully to interaction prediction. The transformer-based interaction module can still learn patterns directly from the primary sequences, although less efficiently than when operating on richer pretrained representations. The observed performance gap, therefore, highlights the complementary contributions of both the pretrained embeddings and the interaction modeling architecture.

6 Conclusion and Future Work

Throughout this work, we studied RNA–protein interaction prediction through the RPIembeddor framework, a Transformer-based architecture operating on frozen embeddings generated by biological foundation models. We first reproduced the original RPIembeddor setup using the RNAInterAct dataset and successfully recovered strong baseline performance. Hyperparameter optimization using NePS with the Hyperband search strategy further improved the reproduced baseline, although this optimization was only performed for the original RNAInterAct setting.

During the RNAInterAct experiments, several observations emerged regarding the influence of dataset design and evaluation protocols. Simple baseline models, especially the MLP, achieved surprisingly strong performance, suggesting that part of the predictive signal was already accessible from pooled frozen embeddings without requiring the full RPIembeddor architecture. In addition, the RNA family-disjoint split produced unstable validation behavior, whereas a relaxed random train-validation split resulted in more stable curves. However, final test performance remained similar across these settings, indicating that validation instability did not necessarily translate into poor generalization.

To further investigate the robustness of RPI prediction models, we introduced an additional dataset based on eCLIP experiments. Compared to RNAInterAct, eCLIP provided substantially larger RNA diversity and experimentally validated binding data. Initial experiments with randomly generated negatives showed that even an RNA-only CNN could achieve unexpectedly high performance, suggesting that the task could be partially solved through sequence-level shortcuts. To address this limitation, we introduced a more biologically meaningful negative generation strategy in which negatives for a given protein were generated from positive interactions of other proteins. This made the task harder and encouraged models to learn protein-specific interaction patterns rather than simply distinguishing binding-like regions from random background sequences.

Using RNAInterAct and eCLIP, we evaluated several RNA and protein foundation models by generating frozen embeddings and training RPIembeddor on top of them.

The results showed that the choice of foundation model affects downstream prediction, but no model was universally optimal across all datasets and splits. ESM-2 remained a strong and reliable protein encoder, especially on RNAInterAct and the more difficult eCLIP protein-disjoint setting. On the RNA side, performance was more dataset-dependent: RNAElectra performed best on RNAInterAct, whereas LAMAR was more competitive on eCLIP. These findings suggest that the usefulness of embeddings is strongly shaped by the downstream dataset, split regime, and interaction task.

Finally, unsupervised embedding quality metrics such as IsoScore, RankMe, NESum, and StableRank were used as complementary tools for representation analysis. These metrics provided useful information about embedding geometry, but they did not fully predict downstream RPIembeddor performance. In particular, RNAElectra achieved strong supervised performance on RNAInterAct despite relatively low unsupervised metric values. This indicates that intrinsic embedding diversity or isotropy alone is insufficient to determine whether an embedding will be useful for a specific interaction prediction task.

Overall, the results of this thesis suggest that improvements in RNA-protein interaction prediction are determined not only by the choice of foundation models, but also by dataset construction, negative generation strategies, split design, and downstream architecture. RNAInterAct was useful for reproducing and validating the original framework, while eCLIP provided a more diverse and experimentally grounded evaluation setting. Across both datasets, foundation model embeddings contributed useful biological signal, but their value was context-dependent rather than universal.

Several directions remain for future work. The current RPIembeddor architecture contains only a single encoder layer, and increasing model capacity through deeper architectures may improve interaction modeling. Alternative pooling strategies, such as mean pooling or attention-based pooling, could also provide richer sequence representations compared to the current first-token pooling approach. Finally, integrating structural or physicochemical properties of RNA and proteins may further improve prediction performance and biological interpretability.

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